Satvik Tripathi

CONTACT INFORMATION

Philadelphia
PA 19104, United States

satvik.tripathi@pennmedicine.upenn.edu
267.939.8668

RESEARCH OBJECTIVES

Academic and clinical research focused on developing clinically relevant AI for radiology, oncology & patient-centered care.

EDUCATION

Doctor of Philosophy, University of Pennsylvania **Graduation: 2031**
Advisor: Dr. James Gee
Medical Physics- Bioengineering (CAMPEP)
Perelman School of Medicine
School of Engineering and Applied Sciences

Mastes in Law, University of Pennsylvania Carey Law School **Graduation: 2030**
Focus: Healthcare & AI
Dean's scholarship

Bachelors of Science, Drexel University **Graduation: 06/2026**
Major: Interdisciplinary Studies
Focus of study: *Computer Science, Public Health, & Medicine*
Presidential Scholar
James Whiting Scholar
Nina Henderson Provost Scholar (Year 2022-23)

PROFESSIONAL EXPERIENCE

AI Scientist, RAD-AID International **Feb 2025 - Present**

Working in the leadership team of the RAD-AID Informatics team to develop, monitor, and implement AI-driven solutions in low- and middle-income countries for global radiology outreach.

Radiology: AI Journal, Radiological Society of North America (RSNA) **Aug 2025 – Present**

Trainee Editorial Board Member (Aug 2025 – Present): Selected to serve on the Trainee Editorial Board of *Radiology: Artificial Intelligence*, contributing to peer review, offering trainee perspectives on editorial strategy, and supporting the dissemination of rigorous, reproducible, and clinically meaningful AI research in radiology.

Associate Editor, Radiology AI Podcast (Nov 2025 – Present): Leading and curating the *Hot Takes* series, featuring expert discussions on emerging controversies, innovations, and policy-relevant issues in artificial intelligence for radiology; supporting content development, editorial direction, and audience engagement.

Leadership Roles, Society for Imaging Informatics in Medicine (SIIM) **September 2024 – Present**

Jr. Vice Chair, Membership Committee (September 2024 – Present): Supporting SIIM membership engagement and growth through outreach, recruitment, and fostering professional community.

Machine Learning Education Committee Member (Jul 2025 – Present): Contributing to educational initiatives advancing machine learning knowledge and best practices in imaging informatics.

Global Outreach Committee Member (Septemeber 2024 – Present): Assisting with global health initiatives by promoting sustainable imaging informatics infrastructure and collaboration in low- and middle-income countries.

R&D Intern, Subtle Medical **Summer 2025, Summer 2026**

Exploratory research on visual-language models (VLMs) for medical imaging, contributing to the development of AI-driven solutions for radiology. Design, implement, and evaluate novel VLM architectures, comparing their performance against conventional pixel-based and NLP models.

Research Experience

Perelman School of Medicine at the University of Pennsylvania/Penn Medicine

Penn Center for Cancer Care Innovation (PC3I)**, Fellow* ***June 2025 – Present

Serving as an Innovation Fellow at PC3I, Abramson Cancer Center, focusing on advancing cancer care delivery through health system innovation and the integration of artificial intelligence to improve quality, equity, and patient-centered outcomes.

McBeth Lab, Department of Radiation Oncology**, Researcher* ***November 2023 – Present

Working with Dr. Rafe Mcbeth, with a focus on the development of AI in Radiation Oncology and Medical Physics.

Center for Global and Population Health Research**, Research Scholar* ***March 2023 – Present

Working with Dr. Farouk Dako as a core member of the center with the mission to develop scalable and sustainable infrastructure in low-resource settings to allow for the advancement of health information technology.

Department of Radiology**, Researcher* ***October 2021 – Present

Working with Dr. Tessa Cook at the Center for Practice Transformation, Dr. Farouk Dako, and various other physicians and researchers. Focusing on AI-based applications for medical imaging research and clinical workflow.

MGH/Harvard-MIT Health Sciences and Technology

Athinoula A. Martinos Center for Biomedical Imaging**, Research Assistant* ***April 2023 – Present

Working with the Quantitative Translational Imaging in Medicine Laboratory, led by Christopher P. Bridge, where I utilized my skills in biomedical image analysis to assist with ongoing research projects.

Harvard Medical School

Division of Vascular & Interventional Radiology at MGH**, Research Assistant* ***April 2023 – Present

Working with Dr. Dania Daye at the Mass General Hospital, focusing on the applications of machine learning in vascular radiology scenarios. Working alongside Dr. Daye, gaining clinical experience in interventional radiology.

Zitnik Lab**, Undergraduate Research Fellow* ***April 2022 – February 2023

Working under Dr. Marinka Zitnik at the Machine Learning for Medicine and Science at the Department of Biomedical Informatics. Focusing on implementing Machine learning and Deep Learning techniques, Graph Neural networks specifically, for biological and medical research.

Harvard TH Chan School of Public Health

Harvard Health Innovation Lab**, Research Scholar* ***May 2023 – Oct 2023

Working to establish a global Data Collaborative for Health Systems Performance (DC-HSP). This collaboration will focus on collaborating over global data to address health systems' performance for Cardiovascular Health and ultimately engaging policymakers to develop appropriate policies, responsive health systems, and better cardiovascular health outcomes.

Drexel University

Sparse Coding Lab**, Research Assistant* ***January 2022 – Present

Working under Dr. Edward Kim at the Department of Computer Science. Focusing on the

application of advanced AI techniques for biomedical data. Currently working on AI-based Covid vaccine risk study funded by Bill Melinda Gates Foundation.

Drexel Society of AI**, Researcher & Vice President* ***July 2021 – May 2023
Lead several research projects on application AI, specifically in biology, healthcare, and medicine. Frequently collaborative with Penn Medicine for clinical practices. Developing AI techniques for large-scale multi modal datasets.

Ayaz Neuroergonomic Research Lab**, Research Assistant* ***October 2021 – June 2022
Worked under Dr. Hasan Ayaz at the Cognitive Neuroengineering and Quantitative Experimental Research Collaborative (CoNQuER CollabOrative). Researching on human subjects using near-infrared (NIR)-based functional optical brain imaging (fNIR) and electroencephalograms (EEG) for stress monitoring, analysis, and visualization of brains' cognitive response.

GRANTS

Google Org

Google Health AI Grant, *(2025)*
Received a $110k Grant for the year 2025-2056 from the Google Org to build health solutions in bostwana using the MedGemma family of models for radiology-specefic tasks. The grant is awarded through RAD-AID International.

University of Pennsylvania

Dr. Mitchelle D. Schnall Award for Transitional Innovation in Radiology (STIR), *(2025)*
Received a $5000 Research Grant for the year 2025-2056 from the Department of Radiology. Grant supports our work on "Enhancing Radiology Resident Education with an Interactive 'Must-See Cases' Platform and LLM-Based Feedback." The project is mentored by Dr. Tessa Cook.

Drexel Univeristy

Undergraduate Research Mini-Grant, *(2024)*
Received a $2000 Research Grant for the Winter Term 2024 from the Office of Undergraduate Research at Pennoni Honors College. Working on Domain-Specific Transfer Learning in Resource Constraint Settings using Deep Learning Algorithm. The project is mentored by Dr. Edward Kim at the Sparse Coding Lab.

Undergraduate Research Mini-Grant, *(2023)*
Received a $4000 Research Grant for the Winter and Spring Term 2023 from the Office of Undergraduate Research at Pennoni Honors College. Working on Expert-level Detection of Diabetic Retinopathy using Interpretable Deep Learning Algorithm. The project is mentored by Dr. Edward Kim at the Sparse Coding Lab.

Undergraduate Research Mini-Grant, *(2022)*
Received a $1500 Research Grant for the Winter Term 2022 from the Office of Undergraduate Research at Pennoni Honors College. Working on a novel bias detection test for AI-based medical imaging techniques. Working at the Sparse Coding Lab, mentored by Dr. Edward Kim. This project is also supported by Dr. Farouk Dako from Penn Medicine.

HONORS & AWARDS

University of Pennsylvania

Lipman Family Fellowship, *The Wharton School at the University of Pennsylvania (2026-27)*

Dean's Scholarship, *University of Pennsylvania Carey Law School (2026)*

Innovation Fellow, *Penn Center for Cancer Care Innovation (2025)*

Magna cum laude award, *50th Annual Pendergrass Symposium (05/2025)*

National Neurotrauma Society

Hovda Award, *NNS 2024 (05/2024)*

Trainee Award, *NNS 2024 (05/2024)*
Data Blitz Award- Top 20 abstract, *NNS 2024 (05/2024)*

American Association for the Advancement of Science (AAAS)

Featured in Science Magazine, *2024 AAAS Annual Meeting (02/2024)*
Honorable Mention in Poster Competition, *2024 AAAS Annual Meeting (02/2024)*
Travel Grant Award, *2024 AAAS Annual Meeting (02/2024)*
Third Prize in Poster Competition, *2023 AAAS Annual Meeting (03/2023)*
Travel Grant Award, *2023 AAAS Annual Meeting (03/2023)*

Harvard Medical School

Undergraduate Research Fellow, *SIBMI (03/2022)*

Drexel University

James Whitning Scholar *(09/2026)*
Philly Codefest Winner 2024 *(04/2024)*
Nina Henderson Provost Scholar *(10/2022)*
Presidential Scholar *(04/2021)*
Cohort of Honors Program 2021 *(04/2021)*

Conferences & Workshops Organized

Co-Orginizer, *Penn Science Policy Symposium 2026 (Theme: Bioethics & future technology)*	**02/2026**
Co-Orginizer, *Penn Science Policy Symposium 2025 (Theme: global health, equity, & policy)*	**02/2025**
Orginizer, *Penn AI Conference at Penn Engineering*	**04/2024**
Core Organizer – Harvard Site, *Harvard Health Systems Innovation Lab 2023 Hackathon*	**05/2023**
Organizing Committee, *Philadelphia Codefest 2023*	**04/2023**
Co-General Chair, *2023 Drexel AI Symposium (Data Centric AI Theme)*	**05/2022**
Co-General Chair, *2022 Drexel AI Symposium (Medicine Theme)*	**05/2022**
Organizing Committee, *Philadelphia Codefest 2022*	**04/2022**
Mentor, *Drexel Winter AI Workshop*	**10/2021**
Mentor, *Drexel Fall AI Workshop*	**08/2021**

Research Publications

43. ***Satvik Tripathi***, *Kristian Quevada, Shahriar Faghani, Pratham Khandelwal, Don Enwerem, and Tessa S. Cook. "Tools and Architectures for Agentic Systems in Clinical AI and Radiology." European Journal of Radiology Artificial Intelligence (2026): 100131.*
42. *Shawn Lyo,* ***Satvik Tripathi***, *Ali Tejani, David Vu, Siddhant Dogra, Hamza Alizai, Tessa Cook, "Less Is More: Radiologists Prefer Brief Narratives in LLM-Generated Clinical History Summaries" J Digit Imaging. Inform. med (2026).*
41. ***Satvik Tripathi,*** *Tessa Cook, "A Systems Approach toward AI-Enabled Radiology Dictation for Workflow Efficiency in India" Indian Journal of Radiology & Imaging (2026).*
40. ***Satvik Tripathi,*** *Ali Tejani, Paul Yi, Dania Daye, "Are radiologists the new AI product managers?" European Journal of Radiology, Volume 205 (2026).*
39. ***Satvik Tripathi,*** *Saisha Shetty, Austin Lin, Colin Zhao, Theodore Kim, Don Enwerem, Jacinta Arnold, Shahriar Faghani, Tessa S Cook, "MARC v1: An Open-Source Multi-Agent Framework for Clinical AI Reasoning and Coordination" Arxiv (2026).*
38. ***Satvik Tripathi***, *Mustafa Ege Seker, Kristian Quevada, Ebubechukwu D Enwerem, Pratham Khandelwal, Emine Meltem, Bera Koca, Shahriar Faghani, Jacinta Arnold, Dania Daye, Tessa S. Cook, "SPARC-Rad: A Multimodal Benchmark Dataset and Evaluation Pipeline for Spatial and Anatomical Reasoning in Radiology Vision-Language Models" Arxiv (2026).*

37. ***Satvik Tripathi***, *Don Enwerem, Kevin Song, Kristian Quevada, Jacinta Arnold, Tessa S. Cook, "Check-Support: A Local LLM-Powered Tool for Automated Manuscript Submission Checklist Selection and Completion" Arxiv (2026).*
36. ***Satvik Tripathi***, *Tessa Cook, Woojin Kim, "Agentic AI in Radiology." Radiology (2026).*
35. *Zaki, Hossam A., Allison Brea, Krishnaveni Parvataneni, Hadeel Saab, Christopher P. Bridge,* ***Satvik Tripathi***, *and Dania Daye, "Retrieval-Augmented Language Models for Patient-Centered Periprocedural Anticoagulation in Interventional Radiology." CardioVascular and Interventional Radiology (2026): 1-11.*
34. ***Satvik Tripathi***, *Dana Alkhulaifat, and Tessa S. Cook. "Positive act of reporting negative results in large language model research: a call for transparency." Journal of the American Medical Informatics Association (2026): ocaf221.*
33. ***Satvik Tripathi***, *"Logic Sketch Prompting (LSP): A Deterministic and Interpretable Prompting Method." arXiv preprint arXiv:2512.22258 (2025).*
32. *Coscia, Adrian,* ***Satvik Tripathi***, *and Dania Daye. "Technical Principles." The Radiology AI Handbook, pp. 19-28. Elsevier, 2026.*
31. ***Satvik Tripathi***, *Marel Huisman, and Tessa Cook, "Doing Good Science in the Age of Large Language Models: CARE Framework for Radiology AI" European Journal of Radiology Artificial Intelligence*
30. ***Satvik Tripathi***, *Enwerem Don and Alkhulaifat, Dana and Sukumaran, Rithvik and Chambers, Charles and Levic, Darco and Daye, Dania and Cook, Tessa, "Chameleon Dataset: A Large Multi-Pathology Synthetic Chest CT Radiology Reports Dataset" SSRN*
29. ***Satvik Tripathi***, *Dana Alkhulaifat, Hansel J. Otero, and Tessa S. Cook. "Towards Pediatric Patient-Friendly Education Material Using Generative AI." Journal of the American College of Radiology: JACR, (2025): S1546-1440.*
28. ***Satvik Tripathi***, *Rupal O'Quinn, and Tessa S. Cook. "Cardiothoracic Imaging AI for Cardiac Diseases." Seminars in Roentgenology. WB Saunders, 2025.*
27. ***Satvik Tripathi***, *Dana Alkhulaifat, Shawn Lyo, Rithvik Sukumaran, Bolin Li, Vedant Acharya, Rafe McBeth, and Tessa S. Cook. "A Hitchhiker's Guide to Good Prompting Practices for Large Language Models in Radiology." Journal of the American College of Radiology 22, no. 7 (2025): 841-847.*
26. ***Satvik Tripathi***, *Dana Alkhulaifat, Florence X. Doo, Pranav Rajpurkar, Rafe McBeth, Dania Daye, and Tessa S. Cook. "Development, Evaluation, and Assessment of Large Language Models (DEAL) Checklist: A Technical Report." NEJM AI 2, no. 6 (2025): AIp2401106. .*
25. *Tessa Cook and* ***Satvik Tripathi****. "You Can't Spell Radiology Without AI." In: Shames, J., Zorn, L., Yoon-Flannery, K. (eds) A Radiologist's Path. Springer, Cham.*
24. ***Satvik Tripathi***, *Dana Alkhulaifat, Meghana Muppuri, Ameena Elahi, and Farouk Dako. "Large Language Models for Global Health Clinics: Opportunities and Challenges," Journal of the American College of Radiology (2025).*
23. ***Satvik Tripathi***, *Mutter, L., Muppuri, M., Dheer, S., Garza-Frias, E., Awan, K., Jha, A., Dezube, M., Tabari, A., Bizzo, B.C. and Dreyer, K.J., 2025. "PRECISE framework: Enhanced radiology reporting with GPT for improved readability, reliability, and patient-centered care." European Journal of Radiology, p.112124.*
22. ***Satvik Tripathi***, *Ali Tejani, Tessa Cook, "Beyond Proprietary Models: The Potential of Open-Source Large Language Models in Radiology," Radiology, 2025 315:1.*
21. ***Satvik Tripathi***, *Jay Patel, Liam Mutter, Felix J. Dorfner, Christopher P. Bridge, Dania Daye, "Large language models as an academic resource for radiologists stepping into artificial intelligence research," Current Problems in Diagnostic Radiology, 2024, ISSN 0363-0188.*
20. ***Satvik Tripathi***, *Kyle Mongeau, Dana Alkhulaifat, Ameena Elahi, Tessa S. Cook. "Large Language Models in Health Systems: Governance, Challenges, and Solutions," Academic Radiology, 05 Oct. 2024.*
19. *Elhakim T, Brea AR, Fidelis W, Paravastu SS, Malavia M, Omer M, Mort A, Ramasamy SK,* ***Tripathi S****, Dezube M, Smolinski-Zhao S, Daye D, "PRO-READ IR:Enhanced PROcedural Information READability for Patient-Centered Care in Interventional Radiology with Large Language Models," Journal of the American College of Radiology (2024)*

18. ***Satvik Tripathi***, *Farouk Dako. "The Potential of Large Language Models for Radiology Report Simplification and Translations," Journal of the American College of Radiology, S1546-1440(24)00524-6. 14 Jun. 2024.*
17. ***Satvik Tripathi***, *Kyla Gabriel, Pushpendra Kumar Trupathi, Edward Kim. "Large Language Models Reshaping Molecular Biology and Drug Development," Chemical Biology & Drug Design, 103, e14568.*
16. *Dania Daye, Regina Parker,* ***Satvik Tripathi***, *Meredith Cox, Sebastian Brito Orama, Christopher Bridge, Raul N. Uppot. "Cascade: Context-Aware Data-Driven Ai for Streamlined Multidisciplinary Tumor Board Recommendations in Oncology," Cancers. 2024; 16(11):1975.*
15. ***Satvik Tripathi***, *Rithvik Sukumaran, Suhani Dheer, Tessa S. Cook. "Promptwise: Prompt Engineering Paradigm for Enhanced Patient-Large Language Model Interactions Towards Medical Education." Available at SSRN.*
14. ***Satvik Tripathi***, *Rithvik Sukumaran, Tessa S Cook. "Efficient healthcare with large language models: optimizing clinical workflow and enhancing patient care," Journal of the American Medical Informatics Association, 2024, ocad258.*
13. ***E****dward Kim, Lucy Robinson, Isamu Isozaki, Noreen Robertson, Charles B. Cairns,* ***Satvik Tripathi***, *and Vicki Seyfert-Margolis. "A Coronavirus Cohort Case Study-Dataset Trends using Machine Learning Methods," 2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), pp. 4213-4219. IEEE, 2023.*
12. ***Satvik Tripathi***, *Azadeh Tabari, Arian Mansur, Harika Dabbara, Christopher P. Bridge, Dania Daye, "From Machine Learning to Patient Outcomes: A Comprehensive Review of AI in Pancreatic Cancer," Diagnostics, 14, no. 2: 174.*
11. ***Satvik Tripathi***, *Kyla Gabriel, Suhani Dheer, Aastha Parajuli, Alisha Augustin, Ameena Elahi, Omar Awan, Farouk Dako, "Understanding Biases and Disparities in Radiology AI Datasets: A Review," Journal of the American College of Radiology, 20.9 (2023): 836-841.*
10. ***Satvik Tripathi***, *Suhani Dheer, Edward Kim "Neuro-Psychological Approaches for Artificial Intelligence." Multidisciplinary Approaches in AI, Creativity, Innovation, and Green Collaboration=IGI Global, 2023. 29-43.*
9. ***Satvik Tripathi***, *Alisha Augustin, Adam Dunlop, Alex Zavalny, Suhani Dheer, Rithvik Sukumaran, Owen Haslam, Edward Kim "Recent Advances and Application of Generative Adversarial Networks in Drug Discovery, Development, and Targeting", Artificial Intelligence in the Life Sciences.*
8. ***Satvik Tripathi***,*Alisha Isabelle Augustin, Farouk Dako, Edward Kim, "Turing Test Inspired Method for Analysis of Biases Prevalent in Artificial Intelligence-Based Medical Imaging," AI Ethics, 2022.*
7. ***Satvik Tripathi***, *Ethan Jacob Moyers, Alisha Augustin, Alex Zavalny, Suhani Dheer, Rithvik Sukumaran, Daniel Schwartz, Farouk Dako, Edward Kim "RadGenNets: Deep learning-based radiogenomics model for gene mutation prediction in lung cancer", Informatics in Medicine Unlocked, 2022, 101062, ISSN 2352-9148,.*
6. ***Satvik Tripathi***, *Alisha Isabelle Augustin, Rithvik Sukumaran, Suhani Dheer, Edward Kim, "HematoNet: Expert level classification of bone marrow cytology morphology in hematological malignancy with deep learning," Artificial Intelligence in the Life Sciences, Volume 2, 2022, 100043, ISSN 2667-3185.*
5. ***Satvik Tripathi***, *Ethan Jacob Moyers, Alisha Augustin, Ansh Aashish Dholakia, Andy Nguyen, Isamu Mclean Isozaki, Daniel Schwartz, Edward Kim "EvoSTS Forecasting: Evolutionary Sparse Time-Series Forecasting."arXiv preprint arXiv:2204.07066.*
4. ***Satvik Tripathi***, *Alisha Augustin, and Edward Kim,"Longitudinal Neuroimaging Data Classification for Early Detection of Alzheimer's Disease using Ensemble Learning Models."TechRxiv.*
3. ***Satvik Tripathi***,*"Early Diagnostic Prediction of Covid-19 using Gradient-Boosting Machine Model."arXiv preprint arXiv:2110.09436.*
2. ***Satvik Tripathi*** *and Thomas Heinrich Musiolik, "Fairness and Ethics in Artificial Intelligence-based Medical Imagining" Ethical Implications of Reshaping Healthcare with Emerging Technologies: 1-16.*
1. ***Satvik Tripathi***, *"Artificial Intelligence: A Brief Review." Analyzing Future Applications of AI, Sensors, and Robotics in Society Book: 1-16.*

Conference Presentations, Abstracts, & Talks

76. *Spatial and Anatomical Reasoning Benchmark for Vision Foundation Models in Radiology , 4th Annual Emory HITI Lab Symposium* ***(Oral Presentation)***
75. *A Clinically Grounded Pipeline Integrating a Temporal Vision Foundation Model and Multi-Agent Report Generation, 4th Annual Emory HITI Lab Datathon* ***(Oral Presentation)***
74. *Radiotion Oncology, Medical Physics, & AI: Roles of Clinician-Scientist, SIBMI, Department of Biomedical Informatics, Harvard Medical School* ***(Invited Lecture)***
73. *MARCv1: Open-Source Multi-Agent Orchestration Framework for Reproducible Clinical AI, 51st Annual Pendergrass Symposium, Department of Radiology, University of Pennsylvania*
72. *Structured Prompting for AI-Assisted Code Generation in Radiology: A Human-in-the-Loop Approach, 51st Annual Pendergrass Symposium, Department of Radiology, University of Pennsylvania*
71. *SPARC-RAD: First Spatial and Anatomical Reasoning Benchmark for Vision Foundation Models in Radiology, 51st Annual Pendergrass Symposium, Department of Radiology, University of Pennsylvania*
70. *Applications of Artificial Intelligence in Interventional Radiology, Clinical Artificial Intelligence in Radiology Categorical Course, ARRS 2026* ***(Invited Lecture)***
69. *Mastering LLM and VLM Prompting Techniques for Imaging Informatics and AI, Imaging Informatics Fellowship, Department of Radiology, University of Pennsylvania* ***(Invited Talk)***
68. *Agentic State of Mind, Imaging Informatics Fellowship, Department of Radiology, University of Pennsylvania* ***(Invited Talk)***
67. *Harmonizing Radiology Workflow and Education through Large Language Models: A Multi-Pronged Implementation Study, Penn AI in Medicine Symposium 2026*
66. *Applications of Artificial Intelligence in Interventional Radiology, ARRS Categorical Course 2026* ***(Invited Lecture)***
65. *Enhanced PROcedural Information TRANSLATability for Patient-Centered Care in Interventional Radiology with Large Language Models (PRO-TRANSLATE IR), SIR 2026*
64. *Retrieval-Augmented Language Models for Patient-Centered Periprocedural Anticoagulation in Interventional Radiology, SIR 2026*
63. *Applications of Artificial Intelligence in Interventional Radiology, WisconsINFORMATICS 2026, Madison, WI 2026*
62. *CheckSupport: An Open-Source LLM Tool for Safer, Reproducible Radiology Research Reporting, 2026 AAR Annual Meeting*
61. *Proof-of-Concept Evaluation of Vision-Language Model into Radiology Workflow: MRI Sequence Standardization for Metadata Harmonization and Scalable Deployment, Radiological Society of North America (RSNA) 2025* ***(Oral Presentation)***
60. *Generating Synthetic Chest CT Radiology Reports Using Large Language Models: A Chameleon Dataset Study, Radiological Society of North America (RSNA) 2025*
59. *From Reports to Interpretations: A Scoping Review of Large Language Models in Radiology, Radiological Society of North America (RSNA) 2025*
58. *Evaluating Large Language Model for Identifying Positive Pulmonary Embolism Cases in Chest CT Reports to Assess Imaging Utilization, Radiological Society of North America (RSNA) 2025*
57. *Harmonizing Radiology Workflow and Education through Large Language Models: A Multi-Pronged Implementation Study, EuSoMII Annual Meeting 2025*
56. *Evaluating Vision-Language Foundation Models for Brain MRI Sequence Classification, Stability, and Metadata Harmonization, SIIM Conference on Artificial Intelligence in Medical Imaging (CAIMI) 2025* ***(Oral Presentation)***
55. *Reasoning-Based Large Language Model Study for Automated Drain Management in Interventional Radiology, SIIM Conference on Artificial Intelligence in Medical Imaging (CAIMI) 2025* ***(Oral Presentation)***
54. *Towards Global Health AI: Strategy, Planning, & Deployment, RAD-AID International Annual Meeting 2025* ***(Invited Talk)***

53. *From Safety to Efficiency and High-Value Imaging: Leveraging Quality Frameworks to Transform Radiology Operations, 2025 ACR Quality and Safety + Informatics Conference* ***(Invited Talk)***
52. *Standardizing LLM Research in Radiology: The Development of the Development, Evaluation, and Assessment of Large Language Models (DEAL) Checklist, 2025 ACR Quality and Safety + Informatics Conference*
51. *Harmonizing Radiology Workflow and Education through Large Language Models: A Multi-Pronged Implementation Study, 50th Annual Pendergrass Symposium, Department of Radiology, University of Pennsylvania*
50. *Frontiers of AI in Medicine and Radiology: Age of LLMs, National Institute of Medical Sciences, Jaipur, India* ***(Invited Lecture)***
49. *Role of AI in Medicine and Global Health, King George's Medical University, Lucknow, India* ***(Invited Lecture)***
48. *Frontiers of Generative AI in Radiology and Patient Care, Department of Radiodiagnosis, King George's Medical University, Lucknow, India* ***(Invited Lecture)***
47. *Development, Evaluation, and Assessment of Large Language Models (DEAL) Checklist: Reporting Methods and Results for LLM-Based Radiology Research, SIIM Annual Meeting 2025* ***(Oral Presentation)***
46. *Hip vs Hype: Critiquing Trending Topics in Large Language Models, SIIM Annual Meeting 2025* ***(Invited Talk)***
45. *Large Language Models as an Affordable and Equitable Solution for Global Health Informatics, SIIM Annual Meeting 2025* ***(Invited Talk)***
44. *DeepSPINE: A Comprehensive Deep Learning Model for Multi-Task Lumbar Spine MRI Analysis, SIIM Annual Meeting 2025*
43. *Can GPT-4 Do Your Systematic Review? Large Language Models as a Research Assistant for Literature Reviews in Radiology, SIIM Annual Meeting 2025*
42. *Chameleon Dataset- Large Synthetic Radiology Report Dataset: A Pilot Study, SIIM Annual Meeting 2025*
41. *Using Open-Source Large Language Models to Extract Labels from Radiology Reports for Machine Learning Applications: A Proof-of-Concept Study, SIIM Annual Meeting 2025*
40. *LLM-Generated Clinical Histories: Evaluating Readability and Efficiency for Radiologic Interpretation, SIIM Annual Meeting 2025* ***(Oral Presentation)***
39. *Pythia: An Integrated Large Language Model Platform for Enhanced Radiology Reporting, SIIM Annual Meeting 2025* ***(Oral Presentation)***
38. *Automating Pennsylvania Act 112 Follow-Up Recommendation Detection in Radiology Reports Using Large Language Models, SIIM Annual Meeting 2025*
37. *Analyzing Trends in Neuroradiology Trainee Discrepancies using Large Language Models, 63rd Annual Meeting of ASNR* ***(Oral Presentation)***
36. *Large Language Models as Affordable and Equitable Solution in Low- and Middle-Income Countries, RAD-AID International Annual Meeting 2024* ***(Oral Presentation)***
35. *Leveraging Psychological Testing for Targeted Cancer Screening Strategies and Identification for At-Risk Population, RAD-AID International Annual Meeting 2024* ***(Oral Presentation)***
34. *Enhancing Cancer Screening Effectiveness through Psychological Testing: Targeted Strategies for Identifying At-Risk Populations, 2024 ACR Quality and Safety + Informatics Conference*
33. *PRECISE Framework and Evaluation for Enhanced Radiology Reporting Using ChatGPT For Patient-Centered Care, 2024 ACR Quality and Safety + Informatics Conference*

    Demographic Reporting In Neuroimaging Studies And Its Effect On Interpreting Results, Radiological Society of North America (RSNA) 2024
32. *DeepSPINE: A Comprehensive Deep Learning Model for Multi-Task Lumbar Spine MRI Analysis, Radiological Society of North America (RSNA) 2024*

31. *Pro-Translate IR: Translation Of Interventional Radiology Procedural Instructions Using Large Language Models To Improve Information Access To Non-English-Speaking Patients, Radiological Society of North America (RSNA) 2024*
30. *Precision Imaging Via Normative Modeling In Neuroimaging: Integrating Large Healthy Control Datasets, Radiological Society of North America (RSNA) 2024*
29. *Benchmarking Optimal Training Data Size For Accurate Classification In Renal Surgical Pathology Reports, Radiological Society of North America (RSNA) 2024*
28. *PRECISE Framework and Evaluation for Enhanced Radiology Reporting Using ChatGPT For Patient-Centered Care, Radiological Society of North America (RSNA) 2024*
27. *GPT-Based Automated Classification and Labeling of Surgical Renal Pathology Reports, 2024 SIIM Conference on Machine Intelligence in Medical Imaging (CMIMI)* ***(Oral Presentation)***
26. *Examining Patient-Large Language Model Interactions Using the PromptWise Paradigm For Medical Education, 2024 SIIM Conference on Machine Intelligence in Medical Imaging (CMIMI)* ***(Oral Presentation)***
25. *PRECISE Framework and Evaluation for Enhanced Radiology Reporting Using ChatGPT For Patient-Centered Care, 2024 Society for Imaging Informatics in Medicine (SIIM24) Conference*
24. *Navigating Bias in AI in Imaging and Medicine, 2024 Society for Imaging Informatics in Medicine (SIIM24) Conference* ***(Session Talk)***
23. *Integration of LLMs in Radiology Clinical Workflow, 2024 Society for Imaging Informatics in Medicine (SIIM24) Conference* ***(Session Talk)***
22. *Improving Patient-Centered Care in Interventional Radiology by using Large Language Models to Enhance Readability level of Patient Procedural Information, 2024 Society for Imaging Informatics in Medicine (SIIM24) Conference* ***(Oral Presentation)***
21. *GPT-4 Self Assessment of Response Readability to Common Questions about Cancer Screening Imaging, 2024 Society for Imaging Informatics in Medicine (SIIM24) Conference* ***(Oral Presentation)***
20. *Deep-Learning-based Outcome Prediction using Diffusion MR in Mild Traumatic Brain Injury Patients, 41st Annual Symposium of the National Neurotrauma Society (NNS 2024)* ***(Oral Presentation)***
19. *Early Detection of Intraductal Papillary Mucinous Neoplasm in the Pancreas using A Multi-channel Deep Convolutional Neural Network, 2024 MIT/MGB AI Cures Conference*
18. *Domain-Specific Transfer Learning for Diabetic Retinopathy Classification in Resource-Constrained Settings, 2024 National Conference on Undergraduate Research.*
17. *Deep Learning-Based Natural Language Processing for Classification of Renal Surgical Pathology Outcomes in a Multi-Site Dataset, 2024 National Conference on Undergraduate Research.*
16. *Transfer Learning For Diabetic Retinopathy Classification In Resource-Constrained Settings, 2024 AAAS Annual Meeting.*
15. *A Coronavirus Cohort Case Study - Dataset Trends using Machine Learning Methods, 2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM).*
14. *PromptWise: Prompt Engineering Paradigm for Enhanced Patient-Large Language Model Interactions Towards Medical Education, Radiological Society of North America 2023.*
13. *Transforming Radiology with Large Language Models: Opportunities and Challenges, Radiological Society of North America 2023.*
12. *CASCADE: Context-Aware Data-driven AI for Streamlined Multidisciplinary Tumor Board Recommendations in Oncology, 2023 ACR Quality and Safety + Informatics Conference.*
11. *ChatGPT Enhanced Radiology Reporting using PRECISE Framework For Patient-Centered Care, 2023 SIIM Conference on Machine Intelligence in Medical Imaging (CMIMI).*
10. *Deep Learning-Based Natural Language Processing for Classification of Renal Surgical Pathology Outcomes in a Multi-Site Dataset, 2023 SIIM Conference on Machine Intelligence in Medical Imaging (CMIMI).*
9. *Deep-Learning Based Brain Age Deviation Index: A Neuroimaging Marker of Neurodegeneration Following mild-TBI, 40th Annual Symposium of the National Neurotrauma Society.*

8. *Deep Learning-based Automated Approaches for Diabetic Retinopathy Detection,Research Grant Awardee, Week of Undergraduate Excellence Drexel 2023.*
7. *Deep learning-based radiogenomics model for gene mutation prediction in lung cancer,2023 Northeast Bioengineering Conference.*
6. *OsteoNet: Deep Learning-based Early Detection of Osteosarcoma in Adolescents, 2023 Northeast Bioengineering Conference.*
5. *Deep learning-based Tumor Tissue Assessment for Osteosarcoma in Adolescents , 2023 AAAS Annual Meeting.*
4. *Deep-perturbations: Modeling Genetic and Chemical Perturbations Using Causally-Inspired GNNs, Harvard Medical School (SIBMI), 2022.*
3. *Turing Test Inspired Method for Analysis of Biases Prevalent in AI-Based Radiology, Research Grant Awardee, Week of Undergraduate Excellence Drexel 2022.*
2. *Going Back to the 1950s to Unveil the Future of AI in Radiology, Drexel AI Symposium, 2022.*
1. *Early Diagnostic Prediction of Covid-19 using Gradient-Boosting Machine Model, Drexel AI Research Conference, 2021.*

Skills

Programming – *Python, R, Bash, Linux, LaTeX*
Frameworks – *TensorFlow, PyTorch, Keras, scikit-learn, Pandas, NumPy/SciPy, ITK/MedPy*
Imaging – *MRI, CT, EEG; DICOM, NIfTI, PACS*
Research – *Clinical Research Methods, Statistical Analysis, Data Harmonization, Reproducibility*
Clinical – *BLS/ACLS, trauma evaluation, hemodynamic monitoring, EPIC EHR, PACS reporting, phlebotomy*

Professional Membership

***Member**, American College of Radiology (ACR)*
***Member**, Radiological Society of North America (RSNA)*
***Member**, Society for Imaging Informatics in Medicine (SIIM)*
***Member**, American Society of Clinical Oncology (ASCO)*
***Member**, National Neurotrauma Society (NNS)*
***Member**, American Association for the Advancement of Science (AAAS)*
***Member**, American Medical Informatics Association (AMIA)*
***Member**, Association of Academic Radiology (AAR)*
***Member**, Radiological Society of North America (RSNA)*
***Member**, Society of Interventional Radiology (SIR)*

References

Dania Daye, MD, PhD
Vice Chair of Practice Transformation
Associate Professor of Radiology
School of Medicine and Public Health
University of Wisconsin
ddaye@wisc.edu

Tessa S. Cook, MD, PhD
Vice Chair of Practice Transformation
Associate Professor of Radiology
Perelman School of Medicine
University of Pennsylvania
tessa.cook@pennmedicine.upenn.edu

Farouk Dako, MD, MPH
Director, Botswana-Upenn Partnership
Associate Professor of Radiology
Perelman School of Medicine
University of Pennsylvania
farouk.dako@pennmedicine.upenn.edu